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FAMILY NAME: _____

GIVEN NAMES: _____

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STUDENT NUMBER:

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SEMESTER 1, 2019 EXAMINATIONS**CITS2200****Physics, Mathematics & Computing****Data Structures and Algorithms**Department of Computer Science & Software
EngineeringThis paper contains: **4 Pages (including title page)**Time Allowed: **2:00** hours**INSTRUCTIONS:**

Total marks for this paper is 50
Students should answer all questions
UWA approved calculators are allowed

THIS IS A CLOSED BOOK EXAMINATION**SUPPLIED STATIONERY****1 x Answer Booklet 18 Pages****ALLOWABLE ITEMS****UWA Approved Calculator with Sticker****PLEASE NOTE**

Examination candidates may only bring authorised materials into the examination room. If a supervisor finds, during the examination, that you have unauthorised material, in whatever form, in the vicinity of your desk or on your person, whether in the examination room or the toilets or en route to/from the toilets, the matter will be reported to the head of school and disciplinary action will normally be taken against you. This action may result in your being deprived of any credit for this examination or even, in some cases, for the whole unit. This will apply regardless of whether the material has been used at the time it is found. Therefore, any candidate who has brought any unauthorised material whatsoever into the examination room should declare it to the supervisor immediately. Candidates who are uncertain whether any material is authorised should ask the supervisor for clarification.

Candidates must comply with the Examination Rules of the University and with the directions of supervisors.

No electronic devices are permitted during the examination.

All question papers and answer booklets are the property of the University and remain so at all times.

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Q1.

(a) Write an appropriate class in Java for implementing a **priority queue** data structure using a linked list. It is enough to explain the variables and the method definitions (no need to write the bodies of the methods).

(5)

(b) Write an *insertion* method for a **priority queue** in Java using the class you have written above. Analyse the complexity of this method.

(5)

Q2.

(a) A heap can be constructed in $O(n)$ time for n inputs. Design an algorithm for finding the k -th smallest element in an unsorted array of length n that is faster than $O(n \log n)$ asymptotically. Analyse the complexity of your algorithm assuming k as a constant.

(5)

(b) Explain the **mergesort** algorithm clearly. Analyse the complexity of **mergesort**.

(5)

Q3.

(a) Given two functions $f(n)$ and $g(n)$, give a formal definition when we can say $f(n) = O(g(n))$.

(5)

(b)

- Prove that $n \log^2 n$ is $O(n^2)$.
- Prove that $n^3 \log n$ is not $O(n^2)$.

(5)

Q4.

(a) Explain clearly *Kruskal's minimum spanning tree* algorithm. Analyse the complexity of this algorithm.

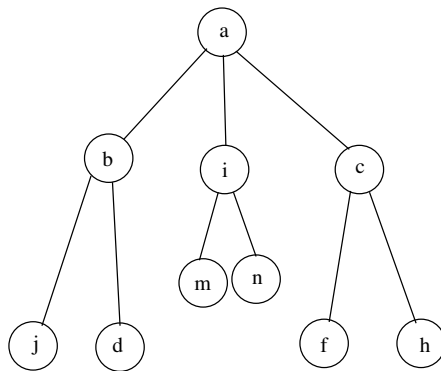
(5)

(b) Explain clearly *Dijkstra's single-source shortest path* algorithm. Analyse the complexity of this algorithm.

(5)

Q5.

(a) Explain clearly the *inorder* and *postorder* traversals of binary trees. Show the ordering of the vertices due to these two traversals on the binary tree given below.



(5)

(b) Explain the *Floyd-Warshall all-pairs shortest path* algorithm. Analyse the complexity of this algorithm.

(5)

—END OF PAPER—